





Management and Curation of EarthPortal Resources

A user interface for the curation of EarthPortal

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1 Introduction

In this short report, we suggest a user interface that could facilitate the curation job on EarthPortal. It builds on the audits produced by our pipeline (SA_Management, v0.3.0), which give, for each ontology, its FAIRness according to O'FAIRe, its compliance with the Metadata Guideline, and a list of suggested actions. Before describing the interface, we answer one question: *who should use it*, the curator (an EarthPortal administrator), the author of an ontology, or both?

2 Development

2.1 Who does the work?

Each suggested action of our audits says who can take it: the **author** (in the ontology file or the submission form), the **curator** (the EarthPortal team), or the **portal** itself (values it generates). For the 68 artefacts of the snapshot of 16 October 2026, the actions of priorities 1 to 3 (Mandatory properties, and properties that cost O'FAIRe points) are split as follows:

Who	Examples of properties	Actions	Share
author	License, Creator, Modification date, Other identifier (DOI), Version IRI	2,626	96 %
curator	Categories, Groups, Curator, Curation date, Included in catalog	81	3 %
portal	CSV dump, Download URL, URI lookup endpoint	37	1 %
	Total	**2,744**	

Table 1: Who can take the actions of priorities 1 to 3, for the 68 artefacts (16 October 2026).

Most of the information that is missing is known only by the authors: the licence they chose, who created the ontology, its DOI. A curator cannot invent it. So an interface used by curators alone would mostly produce messages to authors; and an interface open to anyone would let people change metadata that is not theirs.

*Our answer is **both, with different rights, and in two steps**. First, a curator interface, reserved to EarthPortal administrators. Then, once the suggestions have been validated by the team, an author view limited to the ontologies each author administers.*

OntoPortal already knows these two roles: an **administrator of the portal** can edit every ontology, and each ontology has its own administrators (the field `administeredBy`), who can edit its metadata. The interface should reuse these rights, not create new ones.

	Curator (portal administrator)	**Author (administrator of an ontology)**
Sees	the whole catalogue	only the ontologies they administer
Acts on	the curator actions; the checks of the portal; the requests sent to authors	the author actions of their own ontologies
Decides	accept, edit, reject (with a reason) or postpone each suggestion	fill or correct their metadata, or explain why not
When	step 1	step 2, after the team has validated the suggestions

Table 2: The two roles of the interface.

2.2 Four principles

1. **The pipeline proposes, a person decides.** Nothing is written to the portal without a click. Our checks only see whether a value is *present*, not whether it is *correct*: for example, 55 artefacts have the same default value for their object description properties.
2. **Every decision is recorded**: who, when, which suggestion, and why. A rejected suggestion, with its reason, is not proposed again at the next run. A change can be undone.
3. **Both views of quality are shown, but the score does not decide alone.** Each action says why the guideline asks for it and what it is worth in O'FAIRe. A property such as *Abstract* costs no O'FAIRe point, but the guideline recommends it.
4. **Every result has a date.** The audits describe one snapshot of the catalogue. The interface shows that date, and marks a suggestion as outdated when the value has changed since.

2.3 Three screens

1. **Catalogue**: one row per ontology, with its O'FAIRe and reachable scores, its guideline compliance (for example "Mandatory 16/20, Recommended 2/57"), its open actions, and the date of its last audit. It is our `index.md`, made interactive (sorting, filters).
2. **Ontology**: the two parts of its audit, then its actions grouped by who can take them. For each action, the current value (read live from the portal), the suggestion, the evidence, and the buttons to decide.
3. **Property**: one property of the guideline across the whole catalogue, for example the 23 artefacts without a Creator. This view shows patterns; a group action is still confirmed ontology by ontology.

Figure 1 sketches the second screen, the one where most decisions are taken.

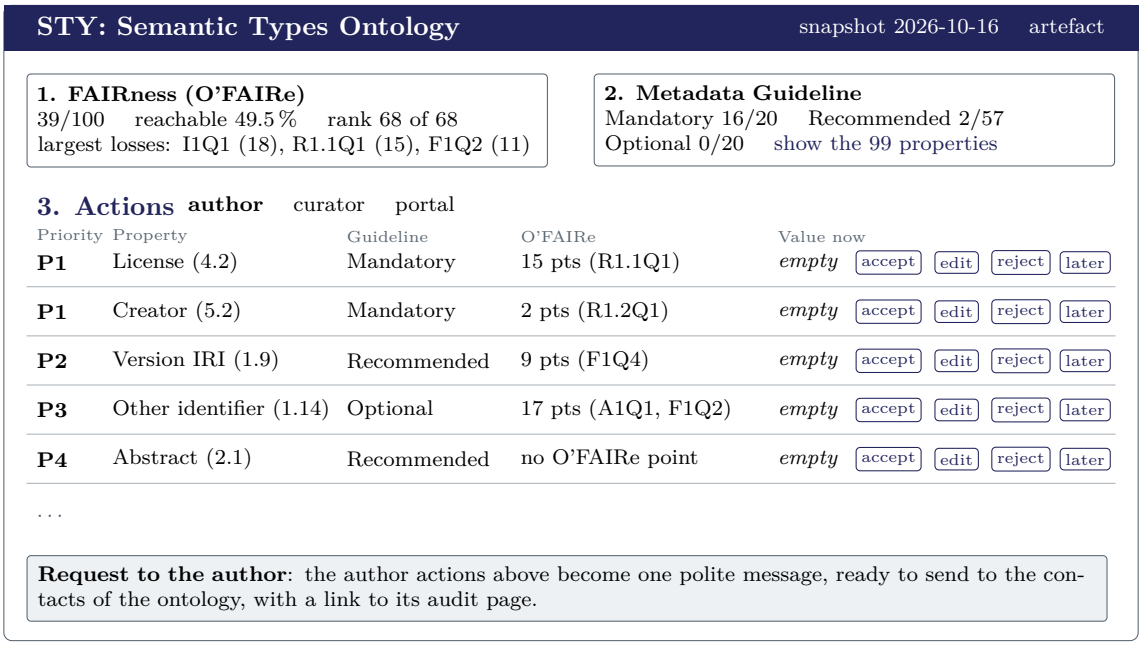


Figure 1: A sketch of the "Ontology" screen, with the data of STY on 16 October 2026. The buttons *accept* and *edit* write to the portal only for the curator actions; for the author actions, the curator prepares a request instead.

2.4 How the interface gets its data

The interface does not need to run our Python code. Since v0.3.0, the pipeline writes **one JSON file per ontology** (`audits/json/<ACRONYM>.json`); the audit page that people read is written from the same file, so the two always agree. Each action looks like this:

```
{ "priority": 3, "why": "Optional, costing O'FAIRe points", "who_fixes": "author",
  "id": "1.14", "entry": "Other identifier", "usage": "Optional",
  "portal_attribute": "dct:identifier", "questions": ["A1Q1", "F1Q2"], "points_share": 17,
  "note": "For F1Q2, only a registered DOI gives the 11 points; any other valid URL gives 6." }
```

The JSON holds no value of the metadata, and so no name or e-mail address. The interface reads the current values live from the portal API, with the rights of the person who is logged in, and writes the accepted changes through the same API.

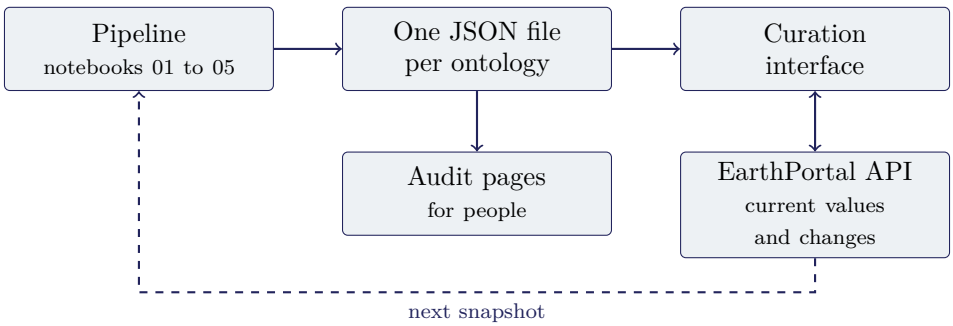


Figure 2: The data flow. The pipeline and the interface only share the JSON files.

2.5 Preventing is better than curing

The submission form of EarthPortal (Figure 3) asks for name, acronym, visibility, administrators, URI, description, representation language, status, location, date of creation and contacts. It does not ask for the **licence**, the **creator** or the **modification date**, and it offers the **categories** without requiring them. These are exactly four of the five Mandatory properties most often empty in the catalogue:

Mandatory property	Artefacts where it is empty	In the submission form
Modification date	27	not asked
Location	24	empty by design when a file is uploaded
Creator	23	not asked
Categories	20	offered, not required
License	15	not asked

Table 3: The Mandatory properties most often empty among the 68 artefacts (16 October 2026).

Asking for these properties at submission time would prevent many actions from ever reaching the curation interface. A small detail on the same form: the text at the top of step 2 ends with an untranslated `%{link}`.

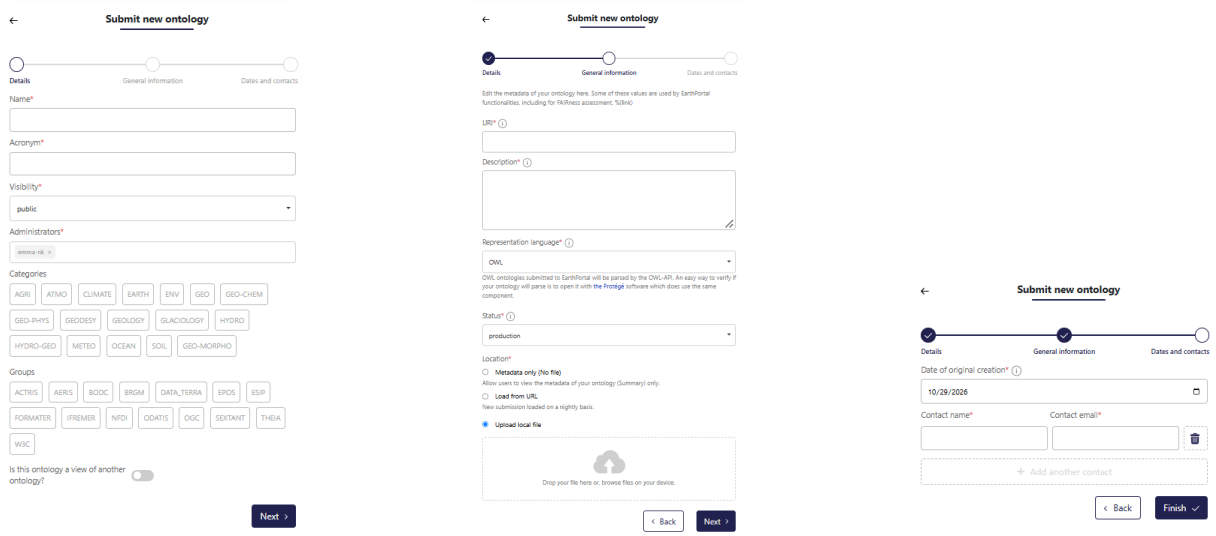


Figure 3: The three steps of the submission form of EarthPortal (production, 29 October 2026).

3 Conclusion

We suggest one curation interface with two roles: first for the EarthPortal curators, then for the authors, each on their own ontologies, using the rights OntoPortal already has. It reads one JSON file per ontology produced by our pipeline, never writes without a person's decision, and keeps a record of every decision. In parallel, adding four properties to the submission form would reduce the work at the source. I would appreciate your feedback, in particular on the two roles and on the place of the interface: inside the EarthPortal administration pages, or as a separate tool.